

ELEC5620 26s1 Final Practice

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W1 Software Architecture Introduction

Q1. Short Answer Semester 2 2023 Final (modified)

A city logistics platform shares route, vehicle and maintenance data with external authorities. List four architecture-level security/privacy risks and propose one modelling or design mitigation for each.

Q2. MCQ

Which statement best distinguishes software architecture from detailed implementation?

- (A) Architecture only describes source-code syntax.
- (B) Architecture captures major structures, responsibilities, connectors and quality trade-offs.
- (C) Architecture is only useful after coding is complete.
- (D) Architecture must avoid stakeholder concerns.

Key focus: Architecture answers should name structures, connectors and quality trade-offs, not only technologies.

Q3. True/False

A single UML class diagram is normally sufficient to justify all architectural quality attributes of a large software-intensive system. Briefly justify your answer.

Pitfall: Do not treat one diagram as the whole architecture; quality attributes normally need several views.

Q4. Short Answer

For a hospital appointment platform, name three stakeholders and one distinct architectural concern for each stakeholder.

Q5. MCQ

A system requires independent scaling of analytics and order processing. Which architectural choice is most directly relevant?

- (A) Merge all logic into one controller class.
- (B) Separate analytics and order processing into components or services with explicit interfaces.
- (C) Remove all traceability links.
- (D) Replace requirements with deployment nodes.

Q6. Short Answer

Explain why the same scenario may require a logical view, process/runtime view and deployment view.

Q7. Table/Diagram

Complete a small table mapping each quality attribute to one design tactic: modifiability, availability, security, performance.

Q8. Short Answer

A student draws every noun in the scenario as a class. Explain two problems with this approach and give a better first-pass modelling heuristic.

Pitfall: Scenario nouns are only candidates; remove vague roles, attributes, duplicated concepts and non-owned external systems.

Q9. Calculation

A two-hour exam has 30 marks and 8 diagram-heavy questions. If a student reserves 10 minutes for reading/checking, estimate a fair maximum writing time per mark and discuss why some marks deserve more drawing time than others.

Q10. Long Question

A smart campus platform manages rooms, sensors, maintenance requests and dashboards.

1. Identify four architecturally significant requirements.
2. Propose three top-level components and their responsibilities.
3. Choose one connector type between two components and justify it.
4. State one trade-off introduced by your architecture.

Key focus: For long design questions, marks usually reward rationale and trade-offs as much as the component list.



W2 Model-Based Software Engineering

Q1. Short Answer Semester 2 2021 Final (modified)

A service provider is onboarded into a home management platform. Describe how a model-based approach can ensure traceability from onboarding requirements to sequence diagram messages and generated interface tests.

Q2. MCQ

In model-driven development, a PIM is best described as:

- (A) A model independent of implementation platform details.
- (B) A screenshot of the final UI.
- (C) A compiler error report.
- (D) A manual test log with no design information.

Key focus: PIM/PSM questions test abstraction level: platform-independent intent vs platform-specific design.

Q3. True/False

Automatic code generation removes the need to validate models. Explain.

Pitfall: Generated artifacts can faithfully implement a wrong or incomplete model; validation moves earlier, it does not disappear.

Q4. Short Answer

Define model, metamodel and model transformation using a simple example from a delivery system.

Q5. MCQ

Which pair is the strongest reason for using MBSE in a complex system?

- (A) More diagrams and fewer stakeholders.
- (B) Traceability and early analysis of design alternatives.
- (C) Avoiding all implementation work.
- (D) Making requirements informal.

Q6. Short Answer

Give two examples of analysis that can be performed on a model before implementation.

Q7. Table/Diagram

Draw a compact chain showing CIM, PIM, PSM and code for a machine-monitoring feature. Label one artifact at each level.

Q8. Short Answer

A generated scheduler behaves incorrectly because the model omitted material-readiness constraints. Is this a tooling problem, modelling problem or both? Explain.

Key focus: Connect modelling omissions to downstream transformation/code behaviour and traceability gaps.

Q9. MCQ

A transformation rule maps every MachineType model element to a concrete adapter class. What must be preserved for maintainability?

- (A) Trace links between source model elements and generated code.
- (B) Random ordering of generated files.
- (C) Unlabelled associations.
- (D) Removal of constraints.

Q10. Long Question

A warehouse robot platform is moving from manual UML diagrams to MBSE.

1. Identify two models that should become authoritative artifacts.
2. Propose one transformation or generation target.
3. State two consistency checks the toolchain should enforce.
4. Explain one limitation of relying on generated artifacts.

Pitfall: Do not claim MBSE eliminates engineering judgement; it shifts judgement into metamodels, constraints and transformations.



W3 Requirements Modelling

Q1. Short Answer Semester 2 2024 Final (modified)

Write a compact use case specification for an importer placing an order through a supply-chain platform, including one flow and one exception.

Q2. MCQ

Which item belongs in a feature diagram rather than a class diagram?

- (A) A mandatory Reporting capability with optional Forecasting.
- (B) A private integer attribute.
- (C) A method return type.
- (D) A constructor signature.

Key focus: Feature diagrams model capability and variability, not implementation fields or method signatures.

Q3. True/False

An optional feature is always less important than a mandatory feature. Explain using a product-line example.

Pitfall: Optional means selectable/variable, not low-value; optional premium or regulated features may be business-critical.

Q4. Short Answer

Rewrite this vague requirement into two verifiable requirements: The system should be fast and secure.

Key focus: Replace vague adjectives with measurable thresholds, actors, data scope and verification method.

Q5. MCQ

A feature group says exactly one of CardPayment, BankTransfer or InvoicePayment may be selected. This is:

- (A) Mandatory decomposition.
- (B) Alternative/XOR group.
- (C) Composition.
- (D) Sequence fragment.

Q6. Short Answer

List the key fields of a use case specification and explain why preconditions differ from postconditions.

Q7. Table/Diagram

Construct a mini feature model for a learning-management notification feature with at least one optional feature and one requires dependency.

Q8. Short Answer

Given the actor Maintenance Technician, propose two user goals and one misuse or exceptional condition.

Q9. MCQ

Which statement is the best cross-tree constraint?

- (A) Analytics requires DataCollection.
- (B) Analytics is blue.
- (C) Order has an ID.
- (D) User clicks button.

Q10. Long Question

For a smart energy platform:

1. Identify five raw ad hoc requirements from the scenario: households monitor usage, retailers publish tariffs, batteries charge automatically, faults alert technicians, and managers view reports.
2. Convert them into a feature hierarchy.
3. Write one use case for automatic battery charging.
4. Identify two non-functional requirements and how they would be verified.

Pitfall: Do not mix feature hierarchy with use-case flow; one is product variability, the other is actor-goal interaction.



W4 Architecture Design and Modelling

Q1. Short Answer Semester 2 2023 Final (modified)

A fleet platform must integrate with an external traffic system during road closures. Propose a sequence of actions and explain one failure-handling mechanism.

Q2. MCQ

Which situation most strongly favours an event-driven architecture?

- (A) Many producers publish asynchronous alerts to several independent consumers.
- (B) One local function call with no integration.
- (C) A static paper form.
- (D) A requirement to hide all events.

Key focus: Event-driven design is mainly about asynchronous decoupling and fan-out, not just using the word event.

Q3. True/False

A layered architecture automatically guarantees high performance. Explain.

Pitfall: Layering improves separation but can add call overhead, latency and rigid dependency paths.

Q4. Short Answer

Compare adapter and strategy patterns in the context of vendor diagnostic systems.

Q5. MCQ

A component diagram should primarily show:

- (A) Components, interfaces and dependencies.
- (B) Only object lifetimes.
- (C) Only feature optionality.
- (D) Only actor goals.

Q6. Short Answer

Give one design tactic for each: auditability, modifiability, scalability.

Q7. Table/Diagram

Sketch a three-component architecture for an inspection-result workflow and label the connector between each component.

Q8. Short Answer

Explain why a single shared database can simplify integration but harm modularity.

Key focus: Mention schema coupling, ownership ambiguity, migration risk and bypassed service contracts.

Q9. MCQ

Which is the best rationale for a plug-in architecture?

- (A) The system must support new machine types without changing the scheduler core.
- (B) The system has no variation points.
- (C) The system has one hard-coded device.
- (D) The system has no interfaces.

Q10. Long Question

A university research platform needs collaboration discovery, project workflows and AI recommendations.

1. Propose a high-level architecture with at least four components.
2. Identify two quality attributes that conflict.
3. Choose one architectural style for integration and justify it.
4. Describe one traceability link from requirement to component.

Pitfall: A style choice without forces and trade-offs reads like memorised terminology, not architecture reasoning.



W5 Modelling Structure

Q1. Short Answer Semester 2 2023 Final (modified)

For an intelligent fleet or smart operations system, identify core classes and relationships needed to model work inspection or maintenance records and external interfaces. Use different relationship types where appropriate.

Q2. MCQ

A WorkOrder is destroyed when its parent ManufacturingOrder is deleted and has no independent lifecycle. The best UML relation is:

- (A) Dependency.
- (B) Composition.
- (C) Realisation.
- (D) Note.

Key focus: Composition is about lifecycle ownership, not just visual containment or importance.

Q3. True/False

Every role in a scenario should become a subclass of User. Explain a better alternative.

Pitfall: Roles are often permissions or assignments; subclass only when behaviour/lifecycle is genuinely different.

Q4. Short Answer

Explain the difference between association, aggregation and composition using a logistics example.

Q5. MCQ

Which multiplicity is most suitable if one machine can produce many sensor readings and each reading belongs to exactly one machine?

- (A) Machine 1 to SensorReading 0..*
- (B) Machine 0..1 to SensorReading 0..1 only
- (C) Machine * to SensorReading *
- (D) No multiplicity needed

Q6. Short Answer

A class diagram has System connected to 18 classes and no other associations. Diagnose the modelling weakness and suggest a repair.

Pitfall: A central God class hides real domain relationships; split responsibilities into services/entities and add direct associations.

Q7. Table/Diagram

The diagram below is incomplete. Add two missing multiplicities, one interface, and one relationship that should be composition.



Q8. Short Answer

Name one place where an interface is more appropriate than inheritance in a smart-home or smart-factory system.

Q9. MCQ

When should an enum often be preferred to a subclass hierarchy?

- (A) For simple finite statuses such as Idle, Running, Fault.
- (B) For unrelated external systems.
- (C) For classes with distinct attributes and behaviour.
- (D) For replacing all associations.

**Q10. Long Question**

For a library equipment-loan system:

1. Identify at least eight candidate classes.
2. Choose two associations and give multiplicities.
3. Identify one inheritance or role-modelling decision and justify it.
4. Add one interface for an external service.

Key focus: High-scoring class answers include multiplicities, relationship semantics and at least one justified abstraction.



W6 Modelling Behaviour

Q1. Short Answer Semester 2 2021 Final (modified)

Design a state machine for a smart security subsystem with normal, alert and maintenance states. Include events, least one recovery transition.

Q2. MCQ

Which diagram is best for showing the lifecycle of a single shipped item?

- (A) State machine diagram.
- (B) Feature diagram.
- (C) Package diagram only.
- (D) Deployment diagram only.

Key focus: Choose state machines when the exam asks about legal states and event-triggered transitions of one object.

Q3. True/False

An activity diagram and a sequence diagram show exactly the same information. Explain.

Pitfall: Activity diagrams show control/workflow; sequence diagrams show ordered messages between lifelines.

Q4. Short Answer

List four essential elements of a good activity diagram for fault handling.

Q5. MCQ

In a sequence diagram, an alt fragment is mainly used for:

- (A) Alternative interaction paths such as success/failure.
- (B) Class inheritance.
- (C) Feature optionality only.
- (D) Multiplicity labels.

Q6. Short Answer

Give three valid state names and three valid transition events for a maintenance ticket.

Q7. Table/Diagram

Construct a numbered communication flow for a quality module receiving an out-of-tolerance inspection result.

Q8. Short Answer

A student labels states as validate data, send request and update dashboard. Explain why these are likely actions rather than states.

Pitfall: States should be stable conditions such as Pending or Faulted; actions are short-lived steps on transitions or activities.

Q9. MCQ

Which transition label is best formed?

- (A) sensorFault [recoverable] / createTicket()
- (B) Fault Thing Nice
- (C) maybe running?
- (D) Class inherits state.

Q10. Long Question

A drone-delivery system handles parcel pickup, flight, charging and emergency landing.

1. Draw or describe an activity flow for dispatching one parcel.
2. Draw or describe a state machine for one drone.
3. State two events that trigger transitions.
4. Explain one difference between your two diagrams.

Key focus: Keep the activity centred on the process and the state machine centred on one drone's lifecycle.



W7 Metamodeling and Software Modelling Languages

Q1. Short Answer Semester 2 2024 Final (modified)

Design a small metamodel for configurable research-collaboration types. Include metaclasses, relationships and formedness constraints.

Q2. MCQ

In the MOF stack, a UML class diagram created for a particular library system is usually:

- (A) M1 model.
- (B) M3 meta-metamodel.
- (C) Runtime object only.
- (D) Compiler binary.

Key focus: Remember the stack: runtime objects conform to models; models conform to metamodels; metamodels conform to MOF-like foundations.

Q3. True/False

A metamodel is merely a large example model. Explain.

Pitfall: A metamodel defines the language of valid models; an example model is just one instance in that language.

Q4. Short Answer

Define abstract syntax, concrete syntax and static semantics for a small workflow DSL.

Q5. MCQ

Which is a valid well-formedness constraint for a workflow metamodel?

- (A) Every workflow must have exactly one start node.
- (B) Blue nodes are prettier.
- (C) Users like diagrams.
- (D) All code should be fast.

Q6. Short Answer

Give one reason to design a DSL rather than use general-purpose UML directly.

Q7. Table/Diagram

Make a four-row table with metaclass, key attribute, relation and constraint for an incident-response modelling language.

Q8. Short Answer

Explain conformance in the statement a model conforms to its metamodel.

Key focus: Use type, relation, multiplicity and constraint satisfaction in the answer.

Q9. MCQ

Which pair best fits UML profile extension?

- (A) Stereotype and tagged value.
- (B) Fork and join only.
- (C) Page number and font size.
- (D) Compiler and linker.

Q10. Long Question

A university wants a modelling language for project governance.

1. Identify five metaclasses.
2. Give multiplicities between three pairs.
3. Write two constraints in plain English or OCL-like form.
4. Provide one example model instance conforming to your metamodel.

Pitfall: Do not answer metamodel questions with only ordinary runtime domain objects; specify modelling concepts and validity rules.



W8 Advanced Topics in Design Modelling

Q1. Short Answer Semester 2 2023 Final (modified)

A fleet platform adds a machine-learning route predictor. Describe integration steps covering data collection, model deployment, architectural interaction and performance risk.

Q2. MCQ

Which design pattern best decouples vendor-specific diagnostic protocols from the core scheduler?

- (A) Adapter.
- (B) Hard-coded conditional chain.
- (C) Delete interface.
- (D) Global variable pattern.

Key focus: Adapter questions usually reward separation between stable core interfaces and unstable vendor/API details.

Q3. True/False

Traceability is only useful for documentation, not for impact analysis. Explain.

Pitfall: Trace links are active engineering evidence: they help find affected models, code and tests after a change.

Q4. Short Answer

Describe how model consistency checking can detect an interaction error between a sequence diagram and a component interface.

Q5. MCQ

The Strategy pattern is useful when:

- (A) Multiple interchangeable algorithms such as priority scheduling and energy-aware scheduling are needed.
- (B) No behaviour varies.
- (C) All classes are deleted.
- (D) Multiplicity is unknown.

Q6. Short Answer

Give two examples of design variability in a smart manufacturing platform.

Q7. Table/Diagram

Create a traceability table linking one requirement to one feature, one use case, one component and one test.

Q8. Short Answer

Explain one risk of over-modelling and one risk of under-modelling in a project-based unit.

Key focus: Balance model detail against decisions, verification value and maintenance cost.

Q9. MCQ

A model says an emergency stop feature exists, but no state transition ever enters a safe stopped state. This is primarily a:

- (A) Cross-model consistency problem.
- (B) Typography preference.
- (C) Runtime colour issue.
- (D) Feature diagram success.

Q10. Long Question

A product line supports standard, safety-critical and low-cost machine controllers.

1. Identify three variation points.
2. Choose two patterns or modelling mechanisms to manage variability.
3. State two consistency rules across feature and class/state models.
4. Explain one trade-off between reuse and simplicity.

Pitfall: Variability is not just optional checkboxes; it must connect to architecture, classes, states and tests.



W9 Agile Software Development and MBSE

Q1. Short Answer Semester 2 2024 Final (modified)

An AI agent recommends research collaborators under competing objectives. Explain how an iterative development could validate objectives, collect feedback and reduce ethical risk.

Q2. MCQ

Which statement best describes just-enough modelling in agile MBSE?

- (A) Produce models that support current decisions and can evolve with feedback.
- (B) Ban all diagrams.
- (C) Finish all possible models before any feedback.
- (D) Treat models as decorative slides only.

Key focus: Just-enough modelling means decision-supporting and maintainable, not minimal for its own sake.

Q3. True/False

Agile means no architecture. Explain.

Pitfall: Agile still needs architecture; the difference is iterative validation and adaptation rather than one frozen design phase.

Q4. Short Answer

Map user story, acceptance criterion and use case to each other for a machine fault notification feature.

Q5. MCQ

Which practice best prevents a model from becoming stale?

- (A) Include model updates in the definition of done and review them during iteration.
- (B) Store models separately and never review them.
- (C) Let only final reports mention models.
- (D) Remove traceability.

Q6. Short Answer

Give two reasons a regulated or safety-critical project may need stronger modelling discipline than a small web prototype.

Key focus: Use verification, traceability, auditability, safety cases and stakeholder accountability as reasons.

Q7. Table/Diagram

Create a sprint mini-plan for implementing vendor onboarding: model update, implementation task, test and review activity.

Q8. Short Answer

Explain how a retrospective could improve a team's modelling process.

Q9. MCQ

In Kanban, WIP limits primarily help by:

- (A) Exposing bottlenecks and improving flow.
- (B) Increasing all unfinished work.
- (C) Replacing validation.
- (D) Hiding blocked tasks.

Q10. Long Question

A team builds an inspection-management feature over three sprints.

1. Define three user stories.
2. For each, identify the model artifact that should change.
3. State two acceptance tests.
4. Explain how architecture risk should be handled early.

Pitfall: Do not postpone architecture risk until the final sprint; prototype or model the risky integration early.



W10 Modern Computing Platforms for Software Modelling

Q1. Short Answer Semester 2 2024 Final (modified)

A new supplier or vendor joins a platform. Describe a sequence for registration, capability or service definition, configuration, access approval and activation.

Q2. MCQ

Which model is most directly concerned with physical nodes and communication paths?

- (A) Deployment diagram.
- (B) Feature diagram only.
- (C) Use case text only.
- (D) Enumeration.

Key focus: Deployment questions are about nodes, artifacts, communication paths and trust boundaries.

Q3. True/False

Cloud deployment removes the need to model security boundaries. Explain.

Pitfall: Cloud shifts boundaries; it does not remove identity, network, storage or tenant-isolation concerns.

Q4. Short Answer

For an IoT platform, distinguish telemetry flow from command/control flow.

Q5. MCQ

An edge device may lose network connectivity but must keep safe local operation. Which design concern is central?

- (A) Offline state management and synchronisation.
- (B) Deleting all local state.
- (C) Ignoring failures.
- (D) Removing audit logs.

Q6. Short Answer

Name two platform-specific details that would appear in a PSM but not a PIM.

Q7. Table/Diagram

Sketch an integration chain: device, gateway, message broker, domain service, analytics dashboard. Label one security control.

Q8. Short Answer

Explain why vendor API versioning is a modelling concern, not only a coding concern.

Key focus: Versioning affects interface contracts, adapters, compatibility constraints and deployment decisions.

Q9. MCQ

Which is the best place to isolate a vendor-specific diagnostic protocol?

- (A) Diagnostic adapter implementing a stable interface.
- (B) Every unrelated domain class.
- (C) A handwritten note outside the model.
- (D) The page footer.

Q10. Long Question

A smart-building platform has sensors, mobile users, cloud analytics and maintenance contractors.

1. Propose a deployment model with at least four nodes.
2. Identify two trust boundaries.
3. Describe one failure scenario and fallback behaviour.
4. Explain how the deployment choice affects one UML interaction diagram.

Pitfall: Do not draw cloud as one magic box; show edge/cloud/mobile/external nodes and trust zones.



W11 Software Development Process Models

Q1. Short Answer Semester 2 2021 Final (modified)

For a home-management system, explain which process model would suit evolving device integrations and why risk of your choice.

Q2. MCQ

The V-model is especially associated with:

- (A) Linking development stages to corresponding verification and validation activities.
- (B) Never testing.
- (C) Random feature selection.
- (D) Removing requirements.

Key focus: V-model answers should connect each development artifact to a verification or validation activity.

Q3. True/False

Waterfall is always inappropriate. Explain with a context where it may be reasonable and a context where it may fail.

Pitfall: Process-model questions are contextual; absolute claims usually lose marks.

Q4. Short Answer

Compare waterfall, spiral and agile in terms of handling requirement uncertainty.

Q5. MCQ

Which process is most explicitly risk-driven?

- (A) Spiral model.
- (B) A single unreviewed code dump.
- (C) Pure ad hoc hacking.
- (D) No process.

Q6. Short Answer

Explain how traceability supports change impact analysis in a process model.

Q7. Table/Diagram

Build a two-column table matching process artifacts to purposes: requirements model, architecture model, test model, deployment model.

Q8. Short Answer

Give two signs that a process has become too documentation-heavy for its project risk.

Key focus: Tie process cost to project risk, stakeholder needs, feedback speed and model freshness.

Q9. MCQ

DevOps adds value to MBSE mainly by:

- (A) Feeding operational evidence back into models and delivery decisions.
- (B) Avoiding monitoring.
- (C) Removing all tests.
- (D) Making architecture invisible.

Q10. Long Question

Choose a process model for developing a safety-aware factory system.

1. State project assumptions.
2. Compare two candidate process models.
3. Select one and justify it using risk, feedback and documentation needs.
4. Identify two model artifacts that must be maintained through the process.

Pitfall: A process choice without assumptions is weak; state uncertainty, safety/regulation and stakeholder availability first.



W12 Advanced Topics: Profiles, Agents and Ethics

Q1. Short Answer Semester 2 2024 Final (modified)

Specify a small UML profile or metamodel extension for a configurable platform. Include at least three stereotypes, classes, key attributes, tagged values or constraints.

Q2. MCQ

A UML stereotype is best described as:

- (A) A lightweight extension applied to a UML element.
- (B) A runtime object instance only.
- (C) A replacement for all diagrams.
- (D) A page layout command.

Key focus: Profiles extend UML lightly: stereotype plus base metaclass plus tags plus constraints.

Q3. True/False

A tagged value without any constraint or semantic purpose can still improve a model substantially. Explain.

Pitfall: Decorative tags rarely help; the extension should enforce or communicate domain semantics.

Q4. Short Answer

Define sensors, actuators/actions, environment and performance measure for an AI maintenance agent.

Q5. MCQ

For an agent with hard safety requirements and softer efficiency goals, the best decision strategy is often:

- (A) Hierarchical constraints before optimisation.
- (B) Ignore safety if efficiency improves.
- (C) Random selection.
- (D) Remove human oversight.

Q6. Short Answer

List three ethical or failure risks for an AI research-collaboration recommender and one mitigation for each.

Q7. Table/Diagram

Create a profile table with columns stereotype, base UML metaclass, tagged value and constraint for a smart hospital system.

Q8. Short Answer

Explain the difference between weighted-sum and Pareto-style decision making for competing objectives.

Key focus: Weighted sums collapse objectives into one score; Pareto reasoning keeps non-dominated trade-off options visible.

Q9. MCQ

Which profile constraint is most meaningful?

- (A) Every <<SafetyCriticalEvent>> must be handled by a transition to a safe state or explicit mitigation.
- (B) Every stereotype must be green.
- (C) Every class name must be long.
- (D) Every diagram must contain exactly seven boxes.

Q10. Long Question

A smart hospital wants an AI triage-support agent.

1. Model the agent's inputs, outputs and stakeholders.
2. Choose a decision strategy for balancing urgency, fairness, workload and safety.
3. Propose a UML profile stereotype useful for the design.
4. Identify three failure modes and mitigation strategies.

Pitfall: For AI-agent ethics, pair each risk with a concrete mitigation and a human oversight boundary.



Answer Key

W1. Q1 risks: access control, data leakage, integrity, audit/compliance; mitigations RBAC, encryption, validation. Q2 B. Q3 False; multiple views needed. Q4 examples: patient privacy, clinician availability, admin reporting. Q5 A. Q6 view answers different concern. Q7 acceptable tactics: abstraction/layering, redundancy, RBAC, caching/queuing. Q8 domain concepts by responsibility/lifecycle. Q9 about 110/30 $\approx$ 3.7 min/mark; diagrams vary by complexity. Q10 should name ASRs, components, connector and trade-off.

W2. Q1 trace requirements to sequence messages/tests. Q2 A. Q3 False; wrong model generates wrong code. Q4 model is abstraction; metamodel defines language; transformation maps models. Q5 B. Q6 consistency, simulation, reachability, constraint checking. Q7 both, but source is incomplete model. Q8 A. Q9 should include authoritative models, generation target, consistency checks, limitation.

W3. Q1 actor importer; pre auth/catalog; main select supplier/order/pay/confirm; alt unavailable item; exception payment/auth failure. Q2 A. Q3 False; optional may be premium/safety-context dependent. Q4 measurable latency/access/encryption requirements. Q5 B. Q6 fields plus before/after distinction. Q7 valid hierarchy with requires. Q8 goals receive/close ticket; exception unsafe access. Q9 A. Q10 should convert raw statements to features/use case/NFRs.

W4. Q1 traffic event sequence with validation, recomputation, driver notification, fallback. Q2 A. Q3 False; layers can add latency. Q4 adapter wraps external protocol; strategy swaps algorithms. Q5 A. Q6 audit log, modular interfaces, scaling/queues. Q7 valid three components/connectors. Q8 shared DB eases data exchange but couples schemas. Q9 A. Q10 coherent architecture, qualities, style, trace link.

W5. Q1 strong answer includes ManufacturingOrder composition WorkOrder, Machine hierarchy/interface, InspectionResult, MaintenanceTicket and multiplicities. Q2 B. Q3 False; prefer User with Role/RBAC unless behaviour/lifecycle differs. Q4 association vs weak whole-part vs strong ownership. Q5 A. Q6 split System into services/entities. Q7 order-work order composition 1..*, work order-machine 0..1, adapter interface. Q8 external payment/diagnostics/traffic provider. Q9 A. Q10 classes, multiplicities, role decision, external interface.

W6. Q1 states include Disarmed/Armed/Alert/Maintenance; transitions with events/guards/recovery. Q2 A. Q3 False; activity shows workflow, sequence shows interactions over time. Q4 start/end, actions, decisions, swimlanes/forks, exceptions. Q5 A. Q6 open/assigned/in progress/resolved; create/assign/fix/close. Q7 numbered objects/messages. Q8 actions are transient; states are stable conditions. Q9 A. Q10 dispatch activity, drone states, events, distinction.

W7. Q1 metaclasses such as CollaborationType, GovernanceRule, WorkflowTemplate, Milestone, ResourcePolicy with constraints. Q2 A. Q3 False; metamodel defines language. Q4 abstract syntax concepts, concrete notation, constraints. Q5 A. Q6 DSL fits domain and enables analysis/generation. Q7 valid table. Q8 model elements satisfy metamodel types/relations/constraints. Q9 A. Q10 metaclasses, multiplicities, constraints, instance.

W8. Q1 data collection, training, deployment, integration, monitoring. Q2 A. Q3 False; traceability enables impact analysis. Q4 message must match interface operation and types. Q5 A. Q6 scheduling policies, devices, safety modes. Q7 valid trace chain. Q8 over = waste/stale; under = missed risk. Q9 A. Q10 variation points, patterns, consistency rules, trade-off.

W9. Q1 iterative feedback, objective validation, ethical checks/human oversight. Q2 A. Q3 False; agile still needs architecture. Q4 story maps actor goal; acceptance criterion verifies; use case expands flows. Q5 A. Q6 safety/regulation and certification/traceability. Q7 valid sprint plan. Q8 inspect modelling pain and adjust definition of done/templates. Q9 A. Q10 stories, model changes, tests, early architecture risk.

W10. Q1 sequence: vendor portal, access control, registry, capability model, adapter config, approval, scheduler activation. Q2 A. Q3 False; cloud adds trust zones. Q4 telemetry reports data; command/control changes device behaviour. Q5 A. Q6 cloud region/protocol/container/device OS. Q7 valid chain with auth/encryption/RBAC. Q8 versioning affects interface contracts and compatibility. Q9 A. Q10 nodes, trust boundaries, fallback, effect on interaction.

W11. Q1 likely agile/spiral/incremental for evolving integrations; risk stale interfaces/tooling. Q2 A. Q3 False; stable regulated contexts may suit waterfall/V, changing contexts fail. Q4 waterfall low uncertainty, spiral risk-driven, agile feedback-driven. Q5 A. Q6 changed req traces to models/tests/components. Q7 valid artifact-purpose table. Q8 low value docs, stale models, delayed feedback. Q9 A. Q10 assumptions, comparison, selection, maintained artifacts.

W12. Q1 e.g. <<MachineType>> Class tag capability; <<DiagnosticAdapter>> Component tag protocol; <<SafetyCriticalEvent>> Signal tag severity; constraint safety event handled. Q2 A. Q3 False; decoration alone has little value. Q4 inputs telemetry/tickets; actions recommend/schedule; env factory; performance safety+downtime. Q5 A. Q6 bias/privacy/drift/overtrust plus audit/redaction/monitor/human review. Q7 valid profile table. Q8 weighted sum scalarises; Pareto explores trade-off frontier. Q9 A. Q10 complete agent model, decision strategy, profile and mitigations.